

High-Dimensional Probability and Applications in Data Science

Personal Notes

Mihir Rao

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1 The Curse of Dimensionality and the Monte Carlo Method

High-dimensional probability studies random objects with many coordinates and uses probabilistic methods to solve problems whose deterministic complexity grows rapidly with dimension. The central contrast in this lecture is between having many observations and having many parameters. Classical statistics is often well suited to the former; the latter creates distinctly geometric and computational difficulties.

1.1 Big Data and High Dimension

A data set can be large in two independent ways. It may contain many data points, or each data point may itself have many coordinates. A list of incomes for a large population has many observations but only one coordinate per observation. In contrast, a 100×100 grayscale image is naturally a vector with 10^4 coordinates, one for each pixel. Text, sound, video, genomic data, medical histories, and game records give further examples of high-dimensional observations.

The number of observations often appears linearly in an algorithm or estimator. Dimension is more dangerous: a procedure that resolves every coordinate separately may require a number of operations exponential in the dimension.

Definition 1.1 (Curse of dimensionality). The *curse of dimensionality* is the exponential growth in computational or statistical complexity that can occur as the ambient dimension increases.

1.2 Deterministic Integration on a Grid

Consider numerically approximating the integral of a function $f : [0, 1]^d \rightarrow \mathbb{R}$:

$$I(f) = \int_{[0,1]^d} f(x) dx.$$

In one dimension, a Riemann sum evaluates f on a grid. If the spacing between adjacent points is ε , then the interval needs approximately $1/\varepsilon$ points. Applying the same idea independently to d coordinates produces a Cartesian grid with

$$N \approx \left(\frac{1}{\varepsilon}\right)^d$$

points. Even a moderate resolution becomes infeasible when d is large. For example, ten points per coordinate require 10^d function evaluations.

The same phenomenon is visible geometrically. The unit cube $[0, 1]^d$ has volume 1, while doubling

every side produces a cube of volume

$$\text{vol}([0, 2]^d) = 2^d.$$

There is exponentially more volume to cover as dimension grows. A regular grid pays for this expansion by placing points in every combination of coordinate locations.

Note: A grid is deterministic and evenly spaced, but its tensor-product structure is the source of its exponential size: multiplying the resolution in each coordinate multiplies the total number of points.

1.3 Random Sampling

Probability offers a different approximation. Rather than cover the cube with a regular grid, draw independent random points

$$X_1, \dots, X_n \sim \text{Unif}([0, 1]^d)$$

and average the observed function values.

Definition 1.2 (Monte Carlo estimator). The Monte Carlo estimator of $I(f)$ based on n independent uniform samples is

$$\widehat{I}_n(f) = \frac{1}{n} \sum_{i=1}^n f(X_i).$$

The calculation rests on a few standard facts. If a random vector X has density p on $\mathbb{R}^d$, then

$$\mathbb{E}f(X) = \int_{\mathbb{R}^d} f(x)p(x) dx.$$

Expectation is linear, so $\mathbb{E}(X_1 + \dots + X_n) = \mathbb{E}X_1 + \dots + \mathbb{E}X_n$ without any independence assumption. Variances add when the random variables are independent:

$$\text{Var}(X_1 + \dots + X_n) = \sum_{i=1}^n \text{Var}(X_i).$$

Finally, the strong law of large numbers states that if $Y_1, Y_2, \dots$ are independent and identically distributed with $\mathbb{E}|Y_1| < \infty$, then

$$\frac{1}{n} \sum_{i=1}^n Y_i \longrightarrow \mathbb{E}Y_1 \quad \text{almost surely.}$$

Almost-sure convergence means that the convergence holds with probability one.

1.4 Unbiasedness and Convergence

The density of a uniform random vector X on $[0, 1]^d$ is 1 inside the cube and 0 outside. Consequently,

$$\mathbb{E}f(X) = \int_{[0,1]^d} f(x) dx = I(f).$$

Linearity of expectation then gives

$$\mathbb{E}\widehat{I}_n(f) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}f(X_i) = I(f).$$

Thus the Monte Carlo estimate is unbiased. If $f(X)$ is integrable, the strong law also gives

$$\widehat{I}_n(f) \longrightarrow I(f) \quad \text{almost surely.}$$

The variance determines the rate at which the estimate concentrates around the integral.

Proposition 1.3 (Mean-square error of Monte Carlo integration). *Suppose $X_1, \dots, X_n$ are independent and uniform on $[0, 1]^d$, and $\text{Var}(f(X_1)) < \infty$. Then*

$$\mathbb{E} \left[\left(\widehat{I}_n(f) - I(f) \right)^2 \right] = \frac{\text{Var}(f(X_1))}{n}.$$

Proof. Unbiasedness implies that the mean-square error equals the variance of the estimator. Independence and identical distribution yield

$$\begin{aligned} \mathbb{E} \left[\left(\widehat{I}_n(f) - I(f) \right)^2 \right] &= \text{Var} \left(\frac{1}{n} \sum_{i=1}^n f(X_i) \right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(f(X_i)) \\ &= \frac{\text{Var}(f(X_1))}{n}. \end{aligned}$$

□

If $|f(x)| \leq 1$ throughout the cube, then $\text{Var}(f(X_1)) \leq \mathbb{E}f(X_1)^2 \leq 1$. The root-mean-square error therefore satisfies

$$\left(\mathbb{E} \left[\left(\widehat{I}_n(f) - I(f) \right)^2 \right] \right)^{1/2} \leq \frac{1}{\sqrt{n}}.$$

The bound contains no explicit dependence on d . Achieving root-mean-square error at most ε requires $n \geq \varepsilon^{-2}$ samples, compared with roughly ε^{-d} grid points for coordinate-wise resolution ε . Random sampling replaces an exponential dependence on dimension with the slower but dimension-independent Monte Carlo rate $n^{-1/2}$.

Note: Monte Carlo points do not attempt to cover the cube. They turn integration into estimation of an expectation, so independence controls error through variance rather than geometric coverage.

Source: Roman Vershynin, *High-Dimensional Probability and Applications in Data Science*, Lecture 1, "The Curse of Dimensionality and the Monte Carlo Method," official course page

<https://www.math.uci.edu/~rvershyn/teaching/hdp/hdp.html> and lecture notes

<https://www.math.uci.edu/~rvershyn/teaching/hdp/Lecture1.pdf>.